



IT-SIMPLERA INSTITUTE

Cyber Security Analyst Internship Program

WEEK 03 ASSIGNMENT

Cybersecurity Fundamentals & Security Concepts

Prepared By: Nanum Shahzad

Department: Cyber Security

Supervisor: Senior Cyber Security Analyst

Submission Deadline: 18th July, 2026

Table of Contents

Table of Contents	2
Task 1: Protocols & Port Numbers	3
Purpose of Each Protocol.....	3
Task 2: HTTP vs HTTPS — 10 Websites.....	4
Task 3: Kali Recon	5
Ping Test	5
Nmap Scan — scanme.nmap.org	5
Nmap Scan — Local IP	6
Response Time and TTL Analysis.....	6
Open Ports and Services Summary	6
Riskiest Open Port	7
Bonus — Purpose of the -sV Flag	7
Task 4: URL-to-Page Research	7
1. DNS Lookup.....	7
2. IP Address Resolution	7
3. TCP Three-Way Handshake	8
4. TLS/HTTPS Handshake	8
5. HTTP Request	8
6. Server Response.....	8
7. Browser Rendering	8
Task 5: Networking Cheat Sheet	9
Task 6: Web Enumeration	10
WhatWeb — google.com.....	10
WhatWeb — github.com.....	10
Whois — google.com.....	10
Whois — github.com	12
Interesting Findings	13
References.....	13

Task 1: Protocols & Port Numbers

The table below summarizes the most common network protocols, their assigned port numbers, transport type, purpose, and whether they are considered secure by default.

Protocol	Port	Type	Purpose	Secure?
FTP	21	TCP	File transfer	Insecure
SSH	22	TCP	Secure remote login	Secure
Telnet	23	TCP	Remote login (plaintext)	Insecure
SMTP	25	TCP	Send email	Insecure (unless TLS)
DNS	53	TCP/UDP	Domain name resolution	Insecure (unless DNSSEC)
DHCP	67/68	UDP	Auto IP assignment	Insecure
HTTP	80	TCP	Web traffic	Insecure
POP3	110	TCP	Retrieve email	Insecure
NTP	123	UDP	Time sync	Insecure
IMAP	143	TCP	Retrieve email (sync)	Insecure (unless TLS)
SNMP	161	UDP	Network device management	Insecure (v1/v2)
LDAP	389	TCP	Directory services	Insecure (unless LDAPS)
HTTPS	443	TCP	Encrypted web traffic	Secure
SMB	445	TCP	File/printer sharing	Insecure (v1)
RDP	3389	TCP	Remote desktop	Secure (if patched/NLA on)

Purpose of Each Protocol

- **FTP (21)** — Lets users upload and download files between a client and a server over a network.
- **SSH (22)** — Provides an encrypted channel for remotely logging into and managing another machine.
- **Telnet (23)** — Allows remote command-line access to a device, but sends everything unencrypted.
- **SMTP (25)** — Handles the sending and relaying of email messages between mail servers.
- **DNS (53)** — Translates human-readable domain names into IP addresses that computers use to connect.
- **DHCP (67/68)** — Automatically assigns IP addresses and network settings to devices joining a network.
- **HTTP (80)** — Transfers web page content between a browser and a web server in plain text.
- **POP3 (110)** — Downloads email from a mail server to a local client, usually removing it from the server.
- **NTP (123)** — Synchronizes the clocks of computers across a network to a common time source.
- **IMAP (143)** — Lets an email client access and sync messages that stay stored on the mail server.
- **SNMP (161)** — Used by network administrators to monitor and manage devices like routers and switches.
- **LDAP (389)** — Queries and manages directory information such as user accounts in an organization.

- **HTTPS (443)** — Transfers web page content the same as HTTP but encrypts it using TLS for security.
- **SMB (445)** — Enables file, printer, and resource sharing between computers on a network.
- **RDP (3389)** — Allows a user to remotely view and control the desktop of another computer.

Task 2: HTTP vs HTTPS — 10 Websites

A bash loop was used to test ten popular websites for their HTTP-to-HTTPS redirect behavior:

```
for site in google.com github.com wikipedia.org yahoo.com bing.com duckduckgo.com amazon.com
stackoverflow.com reddit.com neverssl.com; do echo "=== $site ===" curl -sIL http://$site | grep
-Ei "HTTP/|Location" echo "" done
```

Site	Final Protocol	Auto-Redirect HTTP→HTTPS?	Secure? — Explanation
google.com	HTTP (200)	No — redirects www but stays on HTTP in this trace	Not secure in this test. Only redirected root→www on HTTP; no HTTPS upgrade shown here. In practice browsers still get HTTPS via HSTS preload, but this raw curl trace proves the server itself didn't force it.
github.com	HTTPS (200)	Yes	Secure. HTTP request immediately 301'd to HTTPS, then served over HTTP/2 — all traffic encrypted end-to-end.
wikipedia.org	HTTPS (200)	Yes (double redirect)	Secure. Two-step redirect (HTTP→HTTPS root→HTTPS www) but final connection is fully encrypted.
yahoo.com	404 (HTTP)	No redirect observed	Inconclusive/broken. Server returned 404 on plain HTTP root instead of redirecting — worth noting as an anomaly; real yahoo.com works via www subdomain or HTTPS directly.
bing.com	HTTP (200)	No — same as google, stays HTTP	Not secure in this test. Redirected root→www but never left HTTP; no encryption applied in this trace.
duckduckgo.com	HTTPS (200)	Yes	Secure. Immediate HTTP→HTTPS redirect, clean single hop, encrypted connection.
amazon.com	HTTPS but 503	Yes	Redirect logic secure (HTTP→HTTPS→www HTTPS), but final response was 503 Service Unavailable — likely bot/rate-limit protection blocking the curl request, not a real outage.
stackoverflow.com	HTTPS (200)	Yes	Secure. HTTP→HTTPS redirect, then an internal 302 to /questions, final page served encrypted.
reddit.com	HTTPS (200)	Yes	Secure. HTTP→HTTPS root→HTTPS www, standard double redirect, fully encrypted final connection.
neverssl.com	HTTP (200)	No — by design	Not secure. This is a test site built specifically to never upgrade to HTTPS, useful as a control/negative example in this report.

Task 3: Kali Recon

Ping Test

Command: `ping -c 4 google.com`

```
(kali@kali)-[~]
$ ping -c 4 google.com
PING forcesafesearch.google.com (216.239.38.120) 56(84) bytes of data.
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=1 ttl=128 time=48.7 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=2 ttl=128 time=117 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=3 ttl=128 time=42.3 ms
64 bytes from any-in-2678.1e100.net (216.239.38.120): icmp_seq=4 ttl=128 time=60.0 ms

— forcesafesearch.google.com ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3007ms
rtt min/avg/max/mdev = 42.311/67.029/117.125/29.610 ms
```

- Resolved to forcesafesearch.google.com (216.239.38.120)
- 0% packet loss, 4/4 received
- RTT: min 42.3ms / avg 67.0ms / max 117.1ms / mdev 29.6ms
- TTL: 128 — indicates a Windows-based host or intermediate hop normalizing to 128; slightly unusual for Google's typical Linux infrastructure, likely due to routing/anycast.

Nmap Scan — scanme.nmap.org

Command: `nmap -sV -T4 scanme.nmap.org`

```
(kali@kali)-[~]
$ nmap -sV -T4 scanme.nmap.org
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-17 21:41 PKT
Warning: 45.33.32.156 giving up on port because retransmission cap hit (6).
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.22s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 985 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
19/tcp    filtered chargen
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
23/tcp    filtered telnet
25/tcp    filtered smtp
53/tcp    filtered domain
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered msrpc
139/tcp   filtered netbios-ssn
445/tcp   filtered microsoft-ds
514/tcp   filtered shell
593/tcp   filtered http-rpc-epmap
1900/tcp  filtered upnp
3389/tcp  filtered ms-wbt-server
9929/tcp  open  nping-echo Nping echo
31337/tcp open  tcpwrapped
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 169.42 seconds
```

Figure: Nmap version-detection scan output for scanme.nmap.org

Port	State	Service	Version
19	filtered	chargen	—
22	open	ssh	OpenSSH 6.6.1p1 (Ubuntu, outdated)
23	filtered	telnet	—
25	filtered	smtp	—

Port	State	Service	Version
53	filtered	domain	—
80	open	http	Apache 2.4.7 (Ubuntu, outdated)
135/139/445	filtered	msrpc/netbios/smb	—
514	filtered	shell	—
593	filtered	http-rpc-epmap	—
1900	filtered	upnp	—
3389	filtered	ms-wbt-server	—
9929	open	nping-echo	Nping echo (intentional test service)
31337	open	tcpwrapped	—

Nmap Scan — Local IP

Commands: `ip a` (identify local interface) → `nmap -sV -T4 <your_local_ip>`

```
(kali@kali)-[~]
$ nmap -sV -T4 192.168.136.128
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-07-17 21:45 PKT
Nmap scan report for 192.168.136.128
Host is up (0.000012s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
80/tcp    open  http    Apache httpd 2.4.65
Service Info: Host: 127.0.1.1

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 6.98 seconds
```

Port	State	Service	Version
80	open	http	Apache httpd 2.4.65

Response Time and TTL Analysis

The ping test to google.com (resolved as forcesafesearch.google.com, 216.239.38.120) returned 4 replies with 0% packet loss. The round-trip time averaged 67.0 ms, ranging from a minimum of 42.3 ms to a maximum of 117.1 ms, with a jitter (mdev) of 29.6 ms. The TTL value returned was 128, which typically indicates the replying host started at a TTL of 128 (common for Windows-based systems) or that the packet passed through routing that normalized it to this value, rather than the standard 64 TTL seen on most Linux servers.

Open Ports and Services Summary

scanme.nmap.org (45.....56):

Port	State	Service	Version
22	open	ssh	OpenSSH 6.6.1p1 Ubuntu
80	open	http	Apache httpd 2.4.7 (Ubuntu)
9929	open	nping-echo	Nping echo
31337	open	tcpwrapped	—

Local IP (192.1.....):

Port	State	Service	Version
80	open	http	Apache httpd 2.4.65

Riskiest Open Port

The most significant security risk identified is port 22 (SSH) on scanme.nmap.org, running OpenSSH 6.6.1p1. This version was released in 2014 and is considerably outdated, meaning it likely lacks patches for several publicly disclosed CVEs affecting older OpenSSH releases (e.g. user enumeration and authentication-bypass issues found in similar-era builds). An attacker could use the version banner to fingerprint the exact release and search for a matching exploit, or attempt brute-force/credential-stuffing attacks against the service. Port 80 running Apache 2.4.7 carries a similar risk, as older Apache versions have documented vulnerabilities tied to specific modules and configurations. Running outdated software on internet-facing ports significantly increases the attack surface compared to the local scan, where Apache 2.4.65 is a current release with a much smaller known-vulnerability footprint.

Bonus — Purpose of the -sV Flag

The -sV flag enables Nmap's version detection feature. Instead of simply reporting a port as open, Nmap sends a series of probes to the service running on that port and compares the responses against its signature database to determine the exact software name and version (for example, identifying "OpenSSH 6.6.1p1" rather than just "port 22 open"). This is essential during a security assessment because vulnerabilities are usually tied to specific software versions — knowing the precise version allows an analyst to cross-reference it against known CVEs and prioritize which services need patching most urgently.

Task 4: URL-to-Page Research

This section traces every step that occurs between a user typing a URL into the browser and the fully rendered page appearing on screen.

1. DNS Lookup

When a user types a URL such as `https://www.example.com` into the browser and presses Enter, the very first step is translating that human-readable domain name into a machine-usable IP address. The browser first checks its own local DNS cache to see if it has recently resolved this domain. If not found, it checks the operating system's cache, and then the request is forwarded to a configured DNS resolver (often provided by the ISP or a public resolver like Google's 8.8.8.8). This step exists because computers route traffic using IP addresses, not domain names, so this translation is mandatory before any connection can begin.

2. IP Address Resolution

If the DNS resolver does not already have the IP address cached, it performs a recursive lookup on the domain's behalf. It starts by querying a root DNS server, which points it to the appropriate Top-Level Domain (TLD) server (e.g. for .com). The TLD server then directs the resolver to the domain's authoritative nameserver, which holds

the actual DNS records and returns the correct IP address. This resolved IP address is then cached temporarily (based on the record's TTL) and sent back to the browser, completing the domain-to-IP translation.

3. TCP Three-Way Handshake

With the IP address in hand, the browser's operating system establishes a reliable connection to the web server using TCP. This is done through a three-step process known as the three-way handshake: the client sends a SYN (synchronize) packet to request a connection, the server responds with a SYN-ACK (synchronize-acknowledge) packet confirming it is ready, and the client replies with a final ACK (acknowledge) packet. Once this exchange completes, a stable, ordered, and reliable communication channel exists between the browser and the server, ready to carry data.

4. TLS/HTTPS Handshake

If the site uses HTTPS (as most modern sites do), the browser and server perform a TLS handshake immediately after the TCP connection is established, before any actual web content is exchanged. The browser sends a "ClientHello" message listing supported encryption methods; the server replies with a "ServerHello" along with its digital certificate, which the browser verifies against trusted Certificate Authorities. Both sides then perform a key exchange to generate a shared session key, and once this is complete, all further communication between the browser and server is encrypted, protecting the data from interception or tampering.

5. HTTP Request

With a secure (or plain, in the case of HTTP) connection now open, the browser sends an HTTP request to the server, typically a GET request asking for the specific resource identified by the URL (such as the homepage's HTML file). This request also includes headers containing information like the browser type, accepted content formats, and any cookies relevant to the site.

6. Server Response

The web server receives the request, processes it, and generates a response. This response includes an HTTP status code indicating the outcome, most commonly 200 OK for success, along with the requested content, usually the HTML document for the page, and additional headers describing content type, caching rules, and server details. If the resource has moved or an error occurred, the server instead returns codes such as 301 (redirect) or 404 (not found).

7. Browser Rendering

Once the browser receives the HTML response, it begins parsing the document to construct the Document Object Model (DOM), a structured representation of the page's content. As it parses, the browser identifies references to additional resources such as CSS stylesheets, JavaScript files, and images, and fetches each of these with additional requests following the same connection process described above. The browser then applies CSS to style the DOM, executes JavaScript to add interactivity, and finally paints the fully assembled page onto the screen for the user to see and interact with.

Task 5: Networking Cheat Sheet

A consolidated reference sheet combining protocol purpose, associated security risk, and the recommended secure alternative for each service.

Protocol	Port	TCP/UDP	Purpose	Security Risk	Secure Alternative
FTP	21	TCP	File transfer	Sends credentials and data in plaintext	SFTP / FTPS
SSH	22	TCP	Secure remote login	Weak/default passwords can be brute-forced	N/A (already secure)
Telnet	23	TCP	Remote command-line access	Fully unencrypted, credentials exposed	SSH
SMTP	25	TCP	Sending/relaying email	Can be exploited for spam relay, no encryption by default	SMTPS (SMTP over TLS)
DNS	53	TCP/UDP	Domain name resolution	Vulnerable to spoofing/cache poisoning	DNSSEC / DNS-over-HTTPS
DHCP	67/68	UDP	Automatic IP address assignment	Rogue DHCP servers can hijack network config	DHCP snooping (network-level protection)
HTTP	80	TCP	Web page transfer	Data transmitted in plaintext, interceptable	HTTPS
POP3	110	TCP	Retrieve email from server	Credentials and email sent in plaintext	POP3S (POP3 over TLS/SSL)
NTP	123	UDP	Time synchronization	Can be abused for DDoS amplification attacks	NTS (Network Time Security)
IMAP	143	TCP	Sync/access email on server	Plaintext credentials if unencrypted	IMAPS (IMAP over TLS/SSL)
SNMP	161	UDP	Network device monitoring/management	v1/v2 use plaintext community strings	SNMPv3 (with authentication/encryption)
LDAP	389	TCP	Directory service queries	Plaintext binds expose credentials	LDAPS (LDAP over SSL/TLS)
HTTPS	443	TCP	Encrypted web page transfer	Misconfigured/expired certs can weaken trust	N/A (already secure)
SMB	445	TCP	File and printer sharing	SMBv1 vulnerable to exploits (e.g. EternalBlue)	SMBv3 with encryption enabled
RDP	3389	TCP	Remote desktop access	Frequent brute-force/ransomware target if exposed	RDP over VPN + Network Level Authentication

Task 6: Web Enumeration

WhatWeb — google.com

Command: `whatweb google.com`

```

kali@kali:~$ whatweb google.com
http://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[216.239.38.120], RedirectLocation[http://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://google.com [301 Moved Permanently] Country[UNITED STATES][US], HTTPServer[gws], IP[216.239.38.120], RedirectLocation[https://www.google.com/], Title[301 Moved], UncommonHeaders[content-security-policy-report-only,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
http://www.google.com/ [200 OK] Cookies[AEC,NID, __Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[216.239.38.120], Script, Title[Google], UncommonHeaders[content-security-policy-report-only,accept-ch,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]
https://www.google.com/ [200 OK] Cookies[AEC,NID, __Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[216.239.38.120], Script, Title[Google], UncommonHeaders[content-security-policy-report-only,accept-ch,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]

```

Attribute	Details
Web Server	GWS (Google Web Server)
CMS/Framework	None detected (custom Google stack, not a public CMS)
JS Libraries	Script tag detected but no specific library fingerprinted (Google uses heavily obfuscated/proprietary JS)
Security Headers	X-Frame-Options: SAMEORIGIN; X-XSS-Protection: 0 (explicitly disabled — modern browsers rely on CSP instead); Content-Security-Policy-Report-Only (report-only mode, not enforcing); alt-svc (HTTP/3 support); accept-ch (Client Hints)
Cookies	AEC, NID, __Secure-STRP — NID/AEC are tracking/ad-personalization cookies
IP	216.239.38.120
Notes	HTTP → HTTPS redirect confirmed on both bare domain and www

WhatWeb — github.com

Command: `whatweb github.com`

```

kali@kali:~$ whatweb github.com
https://github.com [301 Moved Permanently] Country[UNITED STATES][US], IP[20.207.73.82], RedirectLocation[https://github.com/]
https://github.com [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object], OpenSearch[opensearch.xml], Script[application/json,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub - Changing the world of software], X-Content-Type-Options[no-sniff], X-Frame-Options[deny], X-XSS-Protection[0]
https://github.com/ [200 OK] Content-Language[en-US], Cookies[_gh_sess,octo,logged_in], Country[UNITED STATES][US], Email[you@domain.com], HTML5, HTTPServer[github.com], HttpOnly[_gh_sess,logged_in], IP[20.207.73.82], Open-Graph-Protocol[object], OpenSearch[opensearch.xml], Script[application/json,module], Strict-Transport-Security[max-age=31536000; includeSubdomains; preload], Title[GitHub - Changing the world of software], X-Content-Type-Options[no-sniff], X-Frame-Options[deny], X-XSS-Protection[0]

```

Attribute	Details
Web Server	github.com (custom server banner, not disclosing underlying stack e.g. nginx/Unicorn)
CMS/Framework	None (custom Ruby on Rails app — not shown in banner but publicly known)
JS Libraries	Script[application/json, module] — ES modules used, no specific library named in fingerprint
Security Headers	Strict-Transport-Security: max-age=31536000; includeSubdomains; preload (strong HSTS); X-Frame-Options: deny (stricter than Google's SAMEORIGIN); Content-Security-Policy present (full enforcement, not report-only); X-Content-Type-Options; Referrer-Policy; x-github-request-id (custom tracing header)
Cookies	_gh_sess, octo, logged_in — session/analytics cookies, marked HttpOnly
IP	20.207.73.82
Notes	HTTP → HTTPS redirect confirmed

Comparison takeaway: GitHub's header posture is noticeably stricter than Google's — full CSP enforcement (not report-only) and X-Frame-Options: deny vs Google's SAMEORIGIN.

Whois — google.com

Command: `whois google.com`

```
(kali@kali)~$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-17T17:42:32Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
```

```
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: google.com
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-08-02T02:17:33+0000
Creation Date: 1997-09-15T07:00:00+0000
Registrar Registration Expiration Date: 2028-09-13T07:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact: https://corp.markmonitor.com/domain/ui/abuse-report
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
Domain Status: serverUpdateProhibited (https://www.icann.org/epp#serverUpdateProhibited)
Domain Status: serverTransferProhibited (https://www.icann.org/epp#serverTransferProhibited)
Domain Status: serverDeleteProhibited (https://www.icann.org/epp#serverDeleteProhibited)
Registrant Organization: Google LLC
Registrant Country: US
Registrant Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Name Server: ns2.google.com
Name Server: ns1.google.com
Name Server: ns4.google.com
Name Server: ns3.google.com
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-07-17T17:42:00+0000 <<<

For more information on WHOIS status codes, please visit:
https://www.icann.org/resources/pages/epp-status-codes
```

Field	Value
Registrar	MarkMonitor Inc.
Creation Date	1997-09-15
Expiry Date	2028-09-14
Name Servers	NS1-NS4.GOOGLE.COM (self-hosted DNS)
Registrant	Google LLC (US) — not privacy-shielded, org name is disclosed
Privacy Notes	Registrant/Tech email hidden behind MarkMonitor web form (ICANN Temp Spec redaction), but organization identity is public

Field	Value
DNSSEC	Unsigned

Whois — github.com

Command: whois github.com

```
(kali@kali)-[~]
$ whois github.com
Domain Name: GITHUB.COM
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:32Z
Creation Date: 2007-10-09T18:20:50Z
Registry Expiry Date: 2026-10-09T18:20:50Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: DNS1.P08.NSONE.NET
Name Server: DNS2.P08.NSONE.NET
Name Server: DNS3.P08.NSONE.NET
Name Server: DNS4.P08.NSONE.NET
Name Server: NS-1283.AWSDNS-32.ORG
Name Server: NS-1707.AWSDNS-21.CO.UK
Name Server: NS-421.AWSDNS-52.COM
Name Server: NS-520.AWSDNS-01.NET
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-07-17T17:42:47Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
```

```
The Registry database contains ONLY .COM, .NET, .EDU domains and
Registrars.
Domain Name: github.com
Registry Domain ID: 1264983250_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-09-07T09:16:33+0000
Creation Date: 2007-10-09T18:20:50+0000
Registrar Registration Expiration Date: 2026-10-09T00:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact: https://corp.markmonitor.com/domain/ui/abuse-report
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
Registrant Organization: GitHub, Inc.
Registrant Country: US
Registrant Email: Select Request Email Form at https://domains.markmonitor.com/whois/github.com
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/github.com
Name Server: dns4.p08.nsonet.net
Name Server: ns-421.awsdns-52.com
Name Server: ns-520.awsdns-01.net
Name Server: dns3.p08.nsonet.net
Name Server: dns1.p08.nsonet.net
Name Server: ns-1707.awsdns-21.co.uk
Name Server: dns2.p08.nsonet.net
Name Server: ns-1283.awsdns-32.org
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2026-07-17T17:39:28+0000 <<<

For more information on WHOIS status codes, please visit:
https://www.icann.org/resources/pages/epp-status-codes
```

Field	Value
Registrar	MarkMonitor Inc.
Creation Date	2007-10-09
Expiry Date	2026-10-09

Field	Value
Name Servers	Mix of NSOne (p08.nsone.net) and AWS Route 53 (awsdns-*) — multi-provider DNS redundancy
Registrant	GitHub, Inc. (US) — not privacy-shielded
Privacy Notes	Same MarkMonitor redaction pattern as Google
DNSSEC	Unsigned

Interesting Findings

- **google.com** — Uses exclusively self-hosted name servers (ns1–ns4.google.com). Very few organizations run their own authoritative DNS infrastructure at this scale; it signals in-house DNS/network engineering rather than reliance on a third-party DNS provider, and reduces external attack surface for DNS-based attacks on their zone.
- **github.com** — Domain expiry is coming up 2026-10-09 — under 3 months from now at the time of this scan. Also notable: GitHub splits authoritative DNS across two independent providers (NS1 and AWS Route 53), a deliberate resilience pattern (if one provider suffers an outage or DDoS, the other continues resolving), unlike Google's single self-managed setup.
- Both domains share the same registrar (MarkMonitor), which is common for large enterprises since MarkMonitor specializes in corporate brand/domain protection (anti-hijacking, monitoring for typosquats, etc.) rather than being a budget consumer registrar.

References

- [1] IANA. Service Name and Transport Protocol Port Number Registry. Internet Assigned Numbers Authority. <https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml>
- [2] Postel, J. (1980). RFC 793 — Transmission Control Protocol. Internet Engineering Task Force (IETF). <https://www.rfc-editor.org/rfc/rfc793>
- [3] Rescorla, E. (2018). RFC 8446 — The Transport Layer Security (TLS) Protocol Version 1.3. IETF. <https://www.rfc-editor.org/rfc/rfc8446>
- [4] Mockapetris, P. (1987). RFC 1035 — Domain Names: Implementation and Specification. IETF. <https://www.rfc-editor.org/rfc/rfc1035>
- [5] Nmap Project. Nmap Reference Guide. Nmap.org. <https://nmap.org/book/man.html>
- [6] Nmap Project. scanme.nmap.org — Official Nmap Test Target. <https://nmap.org/>
- [7] WhatWeb. Next Generation Web Scanner — Documentation. GitHub. <https://github.com/urbanadventurer/WhatWeb>
- [8] ICANN. WHOIS and Domain Registration Data Lookup. Internet Corporation for Assigned Names and Numbers. <https://lookup.icann.org/>
- [9] OWASP Foundation. HTTP Security Response Headers Cheat Sheet. <https://cheatsheetseries.owasp.org/cheatsheets/HTTP-Headers-Cheat-Sheet.html>
- [10] Microsoft. Understanding SMB and the EternalBlue Vulnerability (MS17-010). Microsoft Security Response Center. <https://msrc.microsoft.com/>

[11] OpenBSD Project. OpenSSH Release Notes and Security Advisories. <https://www.openssh.com/releasenotes.html>

[12] Kali Linux Documentation. Official Kali Linux Tool Documentation. <https://www.kali.org/tools/>